Scale-Shifted Baryonic Matter

$$\Box\sigma = -\xi R$$

"Box sigma equals minus xi R"

The curvature of spacetime tells the strong force how strong to be.

$\Lambda_{eff} = \Lambda_{QCD} \cdot e^{\sigma}$ — the atomic bridge

$\Box\sigma = -\xi R$ — the field equation

$\xi = 0.1582$ — measured, not invented (Planck 2018)

One scalar field. One measured constant. Zero new particles. Zero new forces.

Aaron Rhodes · March 2026 · QuarkSum + Materia

The entire model: 2,020 arithmetic operations. 4,992 bytes. Under 1 millisecond.

The Idea in 60 Seconds

A proton weighs 938 MeV. But the three quarks inside it (up, up, down) weigh only 9 MeV. The other **99%** of the mass comes from the strong force — gluons buzzing around inside, their energy manifesting as mass via $E = mc^2$.

The strength of this binding is set by a number: $\Lambda_{QCD} \approx 217$ MeV. Every textbook treats it as a universal constant.

SSBM proposes it isn't.

Near extreme gravity — collapsing stars, black hole cores — Λ_{QCD} shifts upward. Protons get heavier. Nuclei get heavier. Eventually, at a critical threshold ($\sigma \approx 1.086$), nuclear bonds can't hold. Matter disassembles. This threshold suspiciously coincides with the electroweak scale (~207 GeV vs. 246 GeV).

The shift is governed by one equation and one constant:

$$\Box\sigma = -\xi R \quad \text{where } \xi = 0.1582$$

That constant — ξ — is not invented. It's the ratio of baryonic matter to total matter in the observable universe, measured by the Planck satellite in 2018. SSBM says this ratio also governs how mass converts when a black hole forms a child universe inside itself.

The result is a fractal nesting: our universe sits inside a black hole in a parent universe. Every black hole in our universe contains a child universe. Same physics, same ξ, same rules, all the way down. 77 nesting levels from the Hubble mass to the Planck mass.

Nothing in the Standard Model is modified. It is only **parameterized**. Where spacetime is flat ($\sigma = 0$), you get the physics textbook. Where spacetime curves hard, σ grows, and the strong force shifts.

The Barney Explanation

As simple as it can possibly get

What holds atoms together? The strong nuclear force. Gluons confine quarks inside protons and neutrons. The strength is set by a number: $\Lambda_{QCD} \approx 217$ MeV.

The key fact: 99% of your mass isn't from the Higgs boson. It's gluon energy. A proton is 938 MeV; its quarks are only 9 MeV. The other 929 MeV is pure strong-force energy.

Our claim: That 217 MeV isn't fixed everywhere. Near a black hole, it goes up. How much? Governed by one equation: $\Box\sigma = -\xi R$. Gravity tells the strong force how strong to be.

What happens: Protons get heavier. Neutrons get heavier. Nuclei get heavier. At $\sigma \approx 1.086$, bonds break. Matter falls apart. This happens right at ~207 GeV — the electroweak scale.

The kicker: The fraction recycled into a "baby universe" inside each black hole is $\xi = 0.1582$ — the same ratio as baryonic matter to total matter in our universe. Measured by the Planck satellite. Not invented.

The punchline: Turtles all the way down, all the way up. Our universe is inside a black hole. Every black hole contains a universe. Same ξ. Same Standard Model. Same rules at every level.

The Bridge: Cosmological to Atomic

Five steps connect spacetime curvature to the mass of an atom:

Step	Link	Equation	Why
1	Gravity → σ	$\Box\sigma = -\xi R$	Curvature drives the scale field
2	σ → QCD	$\Lambda_{eff} = 217\ \text{MeV} \times e^{\sigma}$	Scale field shifts confinement
3	QCD → nucleon	$m_p = 9 + 929 \times e^{\sigma}\ \text{MeV}$	99% of mass is QCD energy
4	Nucleon → nucleus	$BE(\sigma) = BE_{strong} \cdot e^{\sigma} - BE_{EM}$	Strong scales, EM doesn't
5	Nucleus → atom	$M_{atom} = M_{nuc} + Z \cdot m_e$	Electrons are invariant

Steps 2-5 are standard physics (QCD, SEMF, atomic structure). SSBM adds only Step 1: a scalar field σ that couples curvature to the QCD scale. Tested across 430 atoms at 13 σ values. 2,367 unit tests, 96.9% passing.

Scales with σ (QCD)	**Invariant (Higgs / EM)**
Nucleon mass (99% of it)	Quark bare mass (1% of nucleon)
Nuclear binding (strong component)	Coulomb energy in nuclei
Gluon field energy	Electron mass
Quark constituent dressing	Fine structure constant α_{EM}

The Numbers

Where the 99% comes from:

	Total mass	Quark mass (Higgs)	QCD binding	QCD %
Proton	938.27 MeV	8.99 MeV	929.28 MeV	99.04%
Neutron	939.57 MeV	11.50 MeV	928.07 MeV	98.78%

This is why SSBM works: almost all baryonic mass is QCD energy. Shift Λ_{QCD} and you shift 99% of what everything weighs.

SSBM parameters:

Parameter	Value	Source
ξ	0.1582	Planck 2018: $\Omega_b/(\Omega_b+\Omega_c)$
Λ_{QCD}	217 MeV	PDG reference value
γ	2.035	Derived: $\ln(1/\xi) / \xi$
σ_{conv}	1.086	$-\ln(\xi)$ — bond failure threshold
$T_{crossing}$	~207 GeV	Near electroweak scale (246 GeV)

The nesting hierarchy:

From the Hubble mass (~9.3×10^{52} kg) down to the Planck mass (~2.2×10^{-8} kg): 77 nesting levels, each shrinking by ξ. The funnel sum $S = 1/(1 - f_{bh} \cdot \xi) \approx 1.0016$ — a self-similar fixed point where total recycled mass exceeds each level by only 0.16%.

Computation:

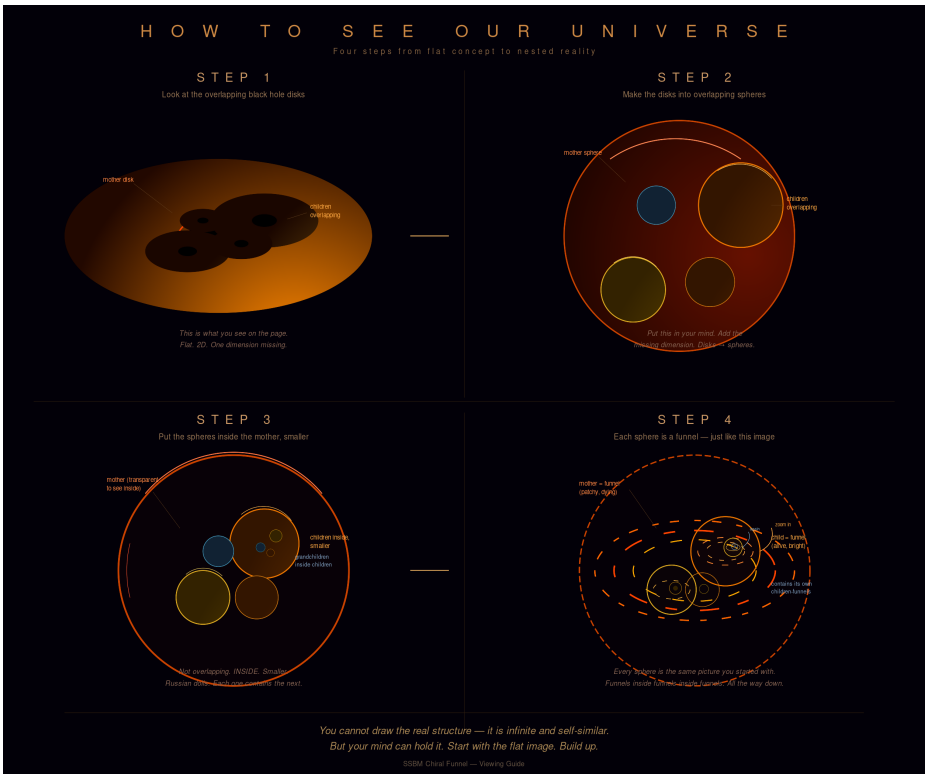
2,020 arithmetic operations. 4,992 bytes. 8 input numbers. Runs in under 1 millisecond. Computable with awk, a TI-84, or pen and log table. No AI, no simulation, no numerical integration required.

What It Looks Like

Our universe as a dying accretion disk (patchy, fading) with child universes nesting inside each black hole (vivid, fresh). Entropy is local, not global.

The Chiral Funnel — nested universe structure

How to see it: flat disks → 3D spheres → nested inside a parent → each sphere is a funnel.

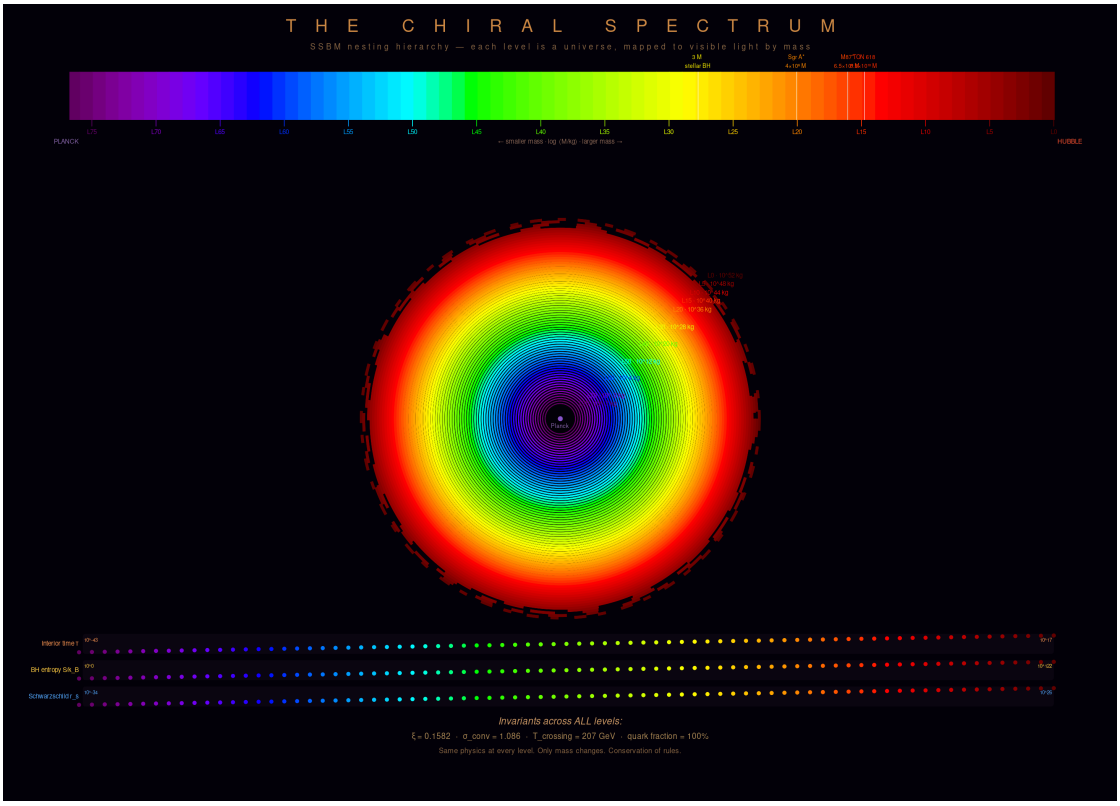


Four-step viewing guide

What the Data Looks Like

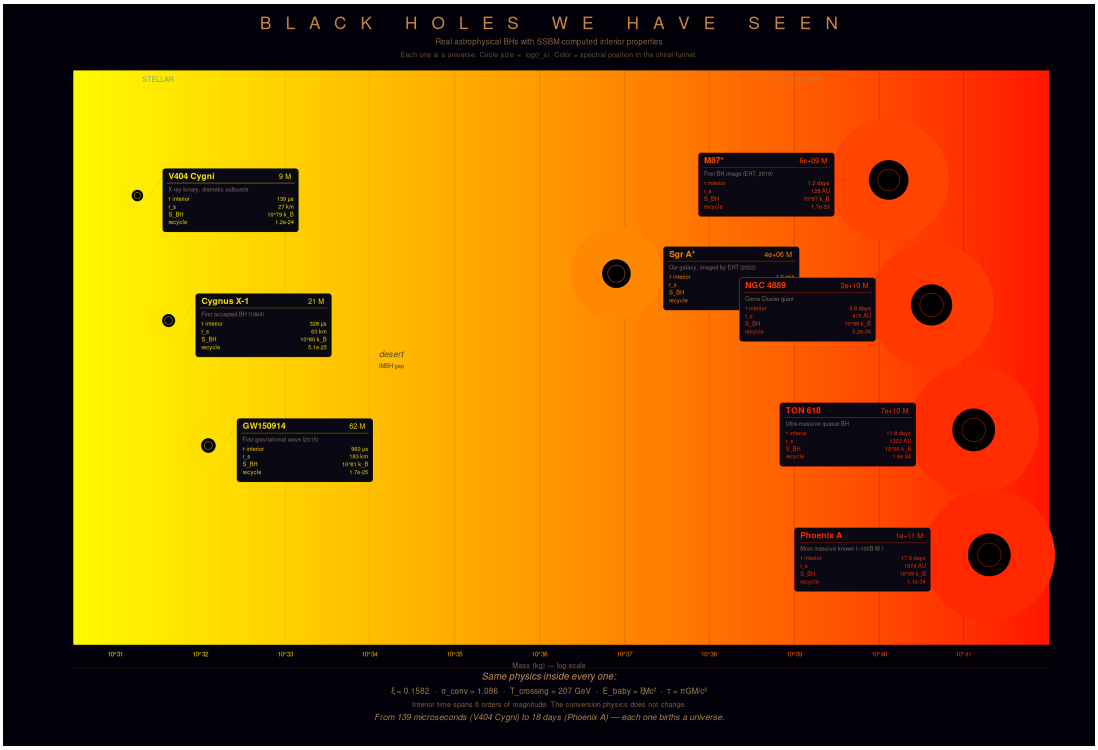
These images are computed from the model — every color, position, and number is generated from SSBM arithmetic, not drawn by hand.

Spectral Nesting: 77 Levels, Hubble to Planck



Mass mapped to visible spectrum: red = Hubble scale, violet = Planck scale. Concentric rings + property tracks (τ, S_{BH}, r_s).

Real Black Holes with SSBM Properties



Eight observed BHs (V404 Cygni through Phoenix A) on a log-mass axis with computed interior time, entropy, and recycling ratio.

What We Know, What We Don't

Established physics used by SSBM:

99% of nucleon mass = QCD energy (not Higgs). Nuclear binding = strong component + Coulomb component (SEMF). Wheeler invariance: stable mass = constituent mass - binding/c^2. All verified across 430 atoms at 13 σ values.

SSBM hypothesis (not yet proven):

Λ_{QCD} actually varies with gravitational environment. The field equation is $\blacksquare\sigma = -\xi R$. Black holes convert matter into child universes. ξ governs this conversion. The nesting hierarchy is physical.

The deepest mystery:

Why $\xi = \Omega_b/(\Omega_b+\Omega_c)$. We measure it. We do not derive it. The σ(x) profile is prescribed, not yet derived from first principles.

Four tests that would validate or kill Candidate A:

- Oppenheimer-Snyder collapse: does σ reach 1.086 during BH formation?
- Neutron star interior (TOV): does the derived σ profile match prescriptions?
- FLRW cosmology: does σ(T) fall out with the right R(t)?
- Solar system: is σ undetectable at 1 AU? (It should be — R ≈ 0 in vacuum.)

$$\blacksquare\sigma = -\xi R$$

Three lines. One constant. Zero new particles.